PCB Manufacturing Process

Single-Layer • Double-Layer • Multilayer

From copper-clad laminate to finished circuit board

Audience: Engineering Students & Junior Engineers

Materials • Imaging • Etching • Lamination • Drilling • Plating • Mask • Finish • Test

Why Understand the Manufacturing Process?

Design decisions you can't make without it

Cost

Every design choice — layer count, via type, finish — changes price.

Yield

Designs that fight the process lose 5-30% yield.

Lead time

Multilayer adds 1-2 weeks vs double-layer turnaround.

DFM

You can't design for manufacturability if you don't know the steps.

Debugging

Spot defects → name them → fix them in the next design iteration.

Career

Hardware engineers who get this are 10× more useful in industry.

What is a PCB?

Substrate + copper + protective layers

- Printed Circuit Board — substrate carrying copper conductors
- Replaces hand-wired point-to-point circuits
- Three core ingredients: substrate, copper, protection (mask + silkscreen)
- Substrate is usually FR-4 (Fibreglass-Reinforced epoxy, type 4)
- Copper layers carry signals + power + ground
- Solder mask protects copper, defines pads
- Silkscreen labels components for assembly

Key dimensions

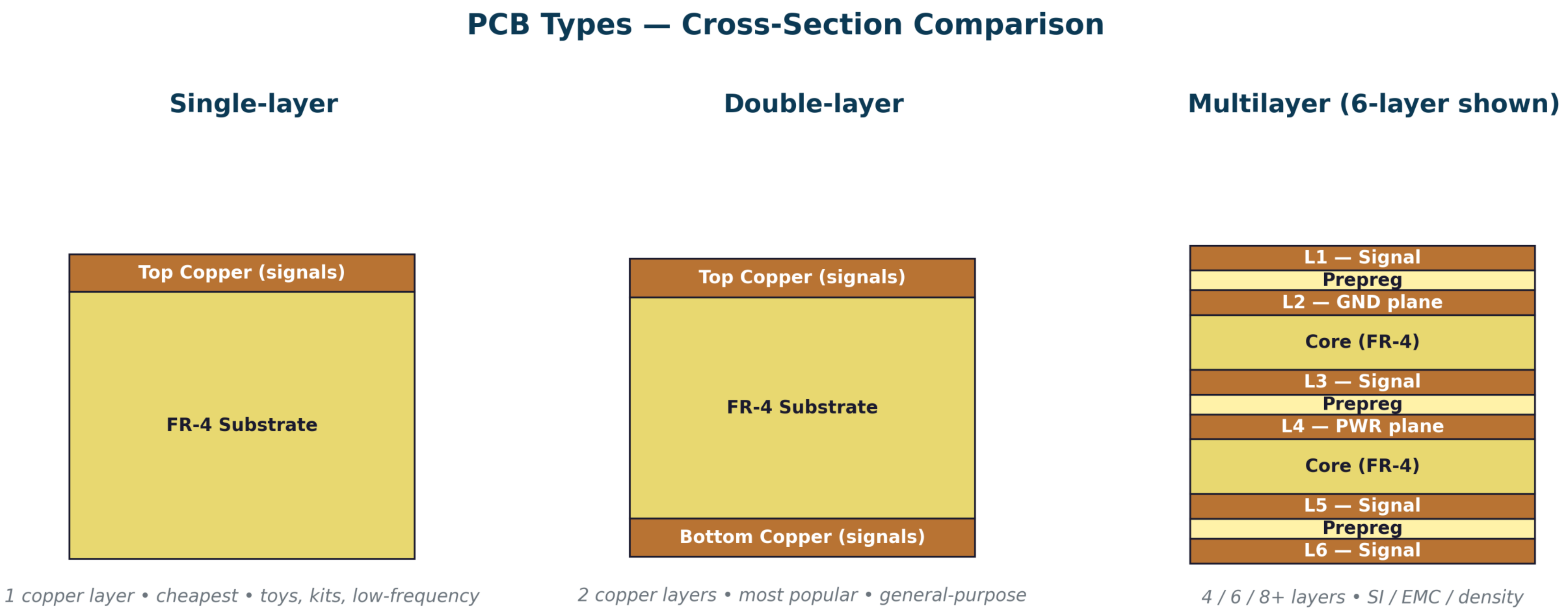
Typical 2-layer FR-4 board: 1.6 mm thick
Standard copper weight: 1 oz/ft² (35 µm thick)
Cu trace width: 6-20 mil (0.15-0.5 mm) typical
Drill holes: 0.3-3 mm

Industry naming

Bare board / Bare PCB → just the substrate + copper (no parts)
PCBA → PCB Assembly (board with components soldered on)
Fab + Assembly → 'turnkey' service from a CM

PCB Types — By Layer Count

Single-layer / Double-layer / Multilayer



Single-Layer PCB

1 copper layer • cheapest tier • limited applications

- Copper foil on one side of the substrate only
- All components and traces on the same side
- No through-hole plating (PTH not needed — no second side)
- Cheapest PCB type — ~₹100-300 per board at low volume
- Simple process: laminate → etch → solder mask → silkscreen → cut
- Limitations: low density, no crossings (need jumpers), low frequency

Typical applications

- LED-driver boards
- Power-supply boards (AC-DC adapter PCBs)
- Toy / kit boards
- Calculator / remote-control PCBs
- Cost-sensitive consumer electronics

When NOT to use

Any design with:

- Multiple ICs that need to interconnect freely
- High-speed signals (clocks, USB, Ethernet)
- Mixed analog + digital sections
- Components on both sides

Double-Layer PCB

2 copper layers • Plated through-holes • Workhorse of the industry

- Copper foil on both top and bottom of the substrate
- Plated through-holes (PTH) connect top ↔ bottom layers
- Components can be placed on both sides (SMD top, THT bottom etc.)
- Most common PCB type for prototypes & consumer products
- Cost: ~₹500-1500 for 50×50 mm at low volume
- Lead time: 3-5 days at fast fabs (JLCPCB / PCBWay)

Typical applications

- Arduino / RPi / ESP32-style hobby boards
- Industrial sensor modules
- Small-batch products
- Education / prototype work
- Most commercial products without high-speed digital

Limitations

- No dedicated power / ground plane (shared GND on bottom)
- EMC harder than 4-layer
- Cross-talk between traces above ~50 MHz
- Need careful manual routing — no internal layers to help

Multilayer PCB

4 / 6 / 8+ copper layers • Dedicated power/ground • SI + EMC king

- Multiple cores (FR-4 with copper) bonded with prepreg
- Inner copper layers dedicated to power and ground planes
- Outer layers for signals — every signal sees a continuous reference plane
- 4-layer is the modern baseline for non-trivial products
- 6 / 8 / 12 layer for dense MCUs, FPGAs, mixed-signal
- BGA packages with > 200 balls usually require ≥ 6 layers

Why more layers?

1. Dedicated ground plane = clean returns + EMC
2. Dedicated power plane = low-impedance distribution
3. Inner signal layers free up outer-layer space
4. Controlled impedance for high-speed traces
5. Higher routing density (BGA escape)

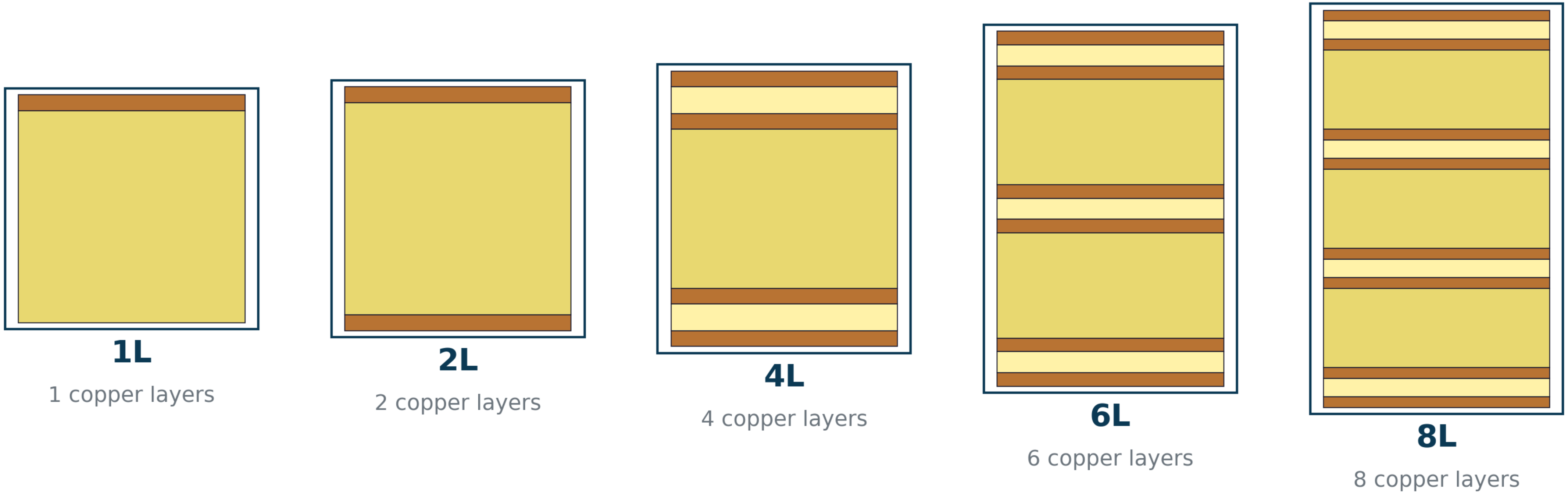
Cost & lead-time impact

4-layer: 2-3× the cost of 2-layer at low volume
6-layer: 4-5×
Lead time: +3-5 days for lamination & extra steps
Often worth it — SI / EMC pain savings > extra cost

Stackup Comparison

1 / 2 / 4 / 6 / 8 layer cross-sections (to scale)

Stackup Comparison — 1 / 2 / 4 / 6 / 8 Layers (to scale)



More layers → better signal/power integrity, denser routing, higher cost.
4-layer is the modern industry baseline for non-trivial products.

Manufacturing Flow — 13 Stages

Single & double-layer skip the multilayer-specific stages (2-4)

PCB Manufacturing Process — 13-Stage Flow

Stages 2-4 apply ONLY to multilayer boards. Single & double-layer skip inner-layer + lamination.

Step 1 — Materials Prep

Substrates, prepregs, copper foils, surface treatments

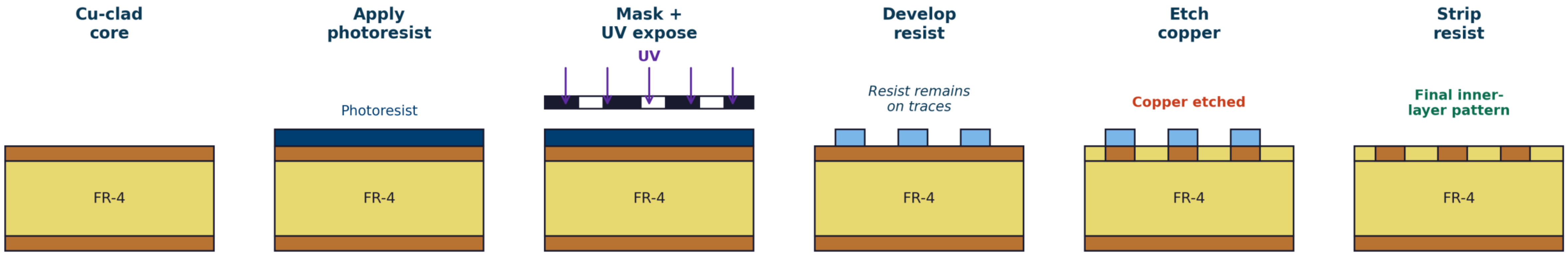
Step 1 — Materials & Substrates

FR-4 Core	Prepreg	Copper foil
Glass-epoxy laminate 0.4–1.6 mm typical Dk ≈ 4.2-4.7	B-stage glass cloth + uncured resin Bonds layers in pressing	Electrodeposited copper ½oz / 1oz / 2oz / heavy (17 / 35 / 70 µm)
Resin systems	**Reinforcement**	**Surface treatments**
Epoxy (FR-4) most common Polyimide (Kapton) for flex Rogers for RF	E-glass fibre weave or non-woven aramid (driving Dk + Tg)	Black/brown-oxide treatment on Cu for prepreg adhesion

Steps 2-3 — Inner-Layer Imaging + Etching

Multilayer ONLY • Photolithography pattern transfer

Inner-Layer Imaging & Etching (Multilayer Only)

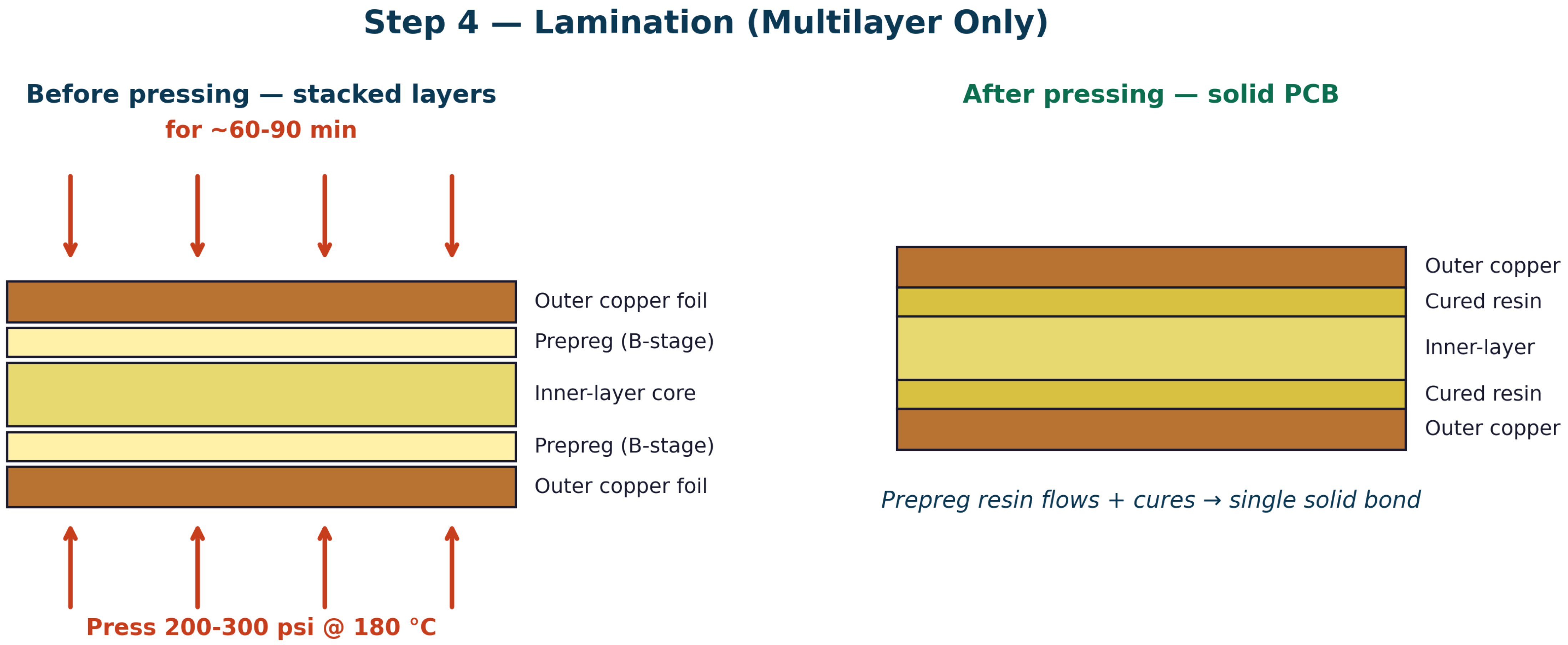


Single & double-layer SKIP this — they have no inner layers to image.

Materials: dry-film photoresist (DFR) laminated to Cu-clad core, UV exposed through phototool, developed in alkaline solution, copper etched in ferric chloride or cupric chloride, resist stripped.

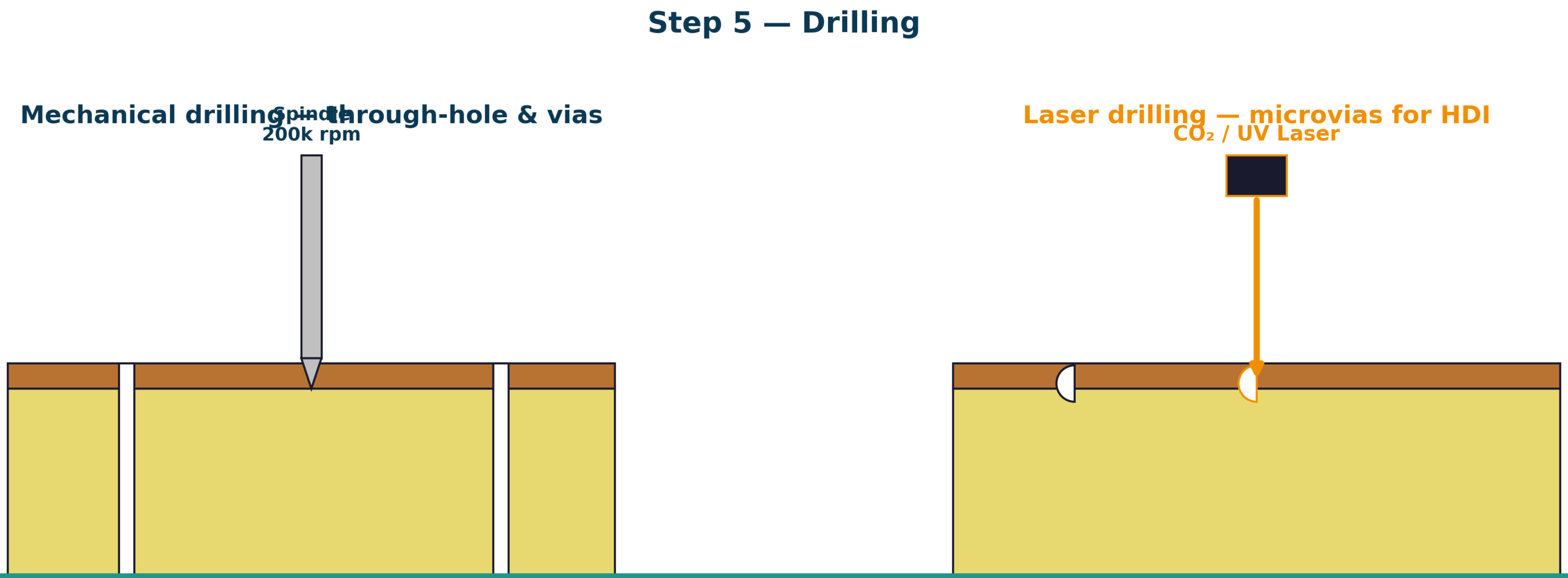
Step 4 — Lamination

Multilayer ONLY • Heat + pressure bonds the stack into one solid PCB



Step 5 — Drilling

Mechanical drills for through-holes & vias • Laser for microvias (HDI)

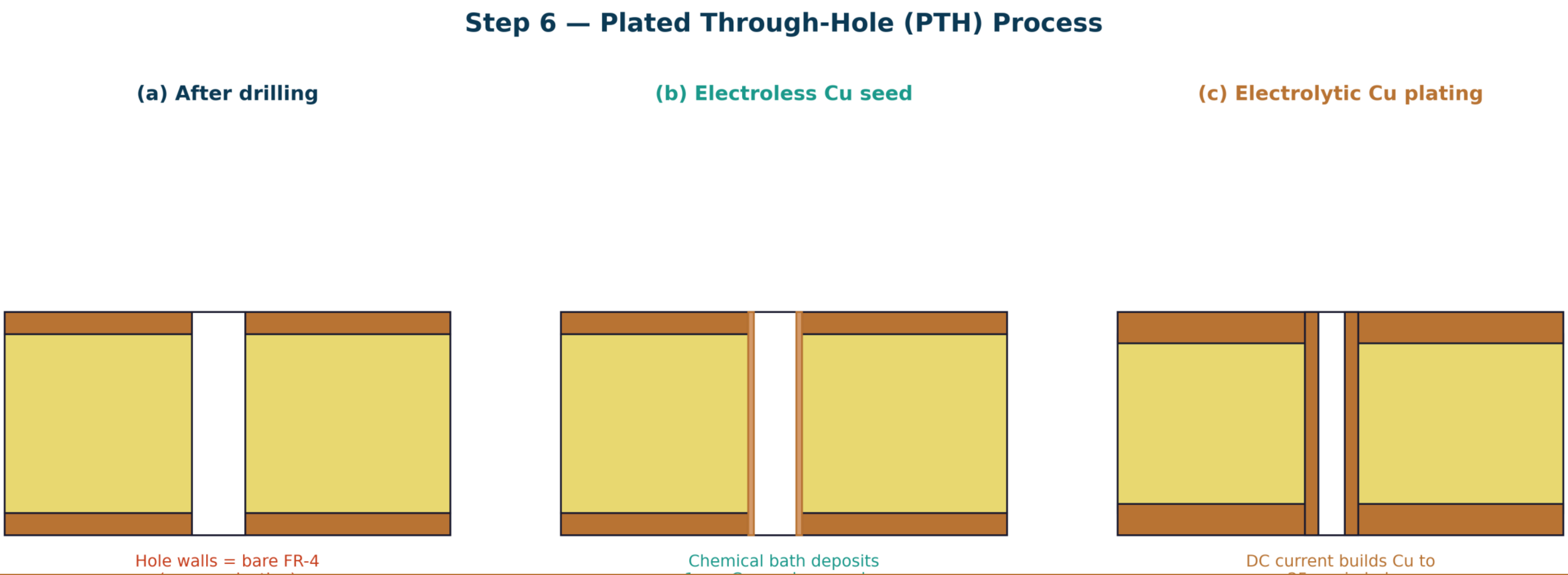


Drilling specs

Standard mechanical: 0.3 mm minimum drill, 0.15 mm annular ring • Laser (UV/CO_2): 50-150 µm microvias for HDI • Aspect ratio (depth/diameter) ≤ 10:1 for reliable plating

Step 6 — Plated Through-Hole (PTH)

Make non-conductive hole walls conductive • Two-step plating

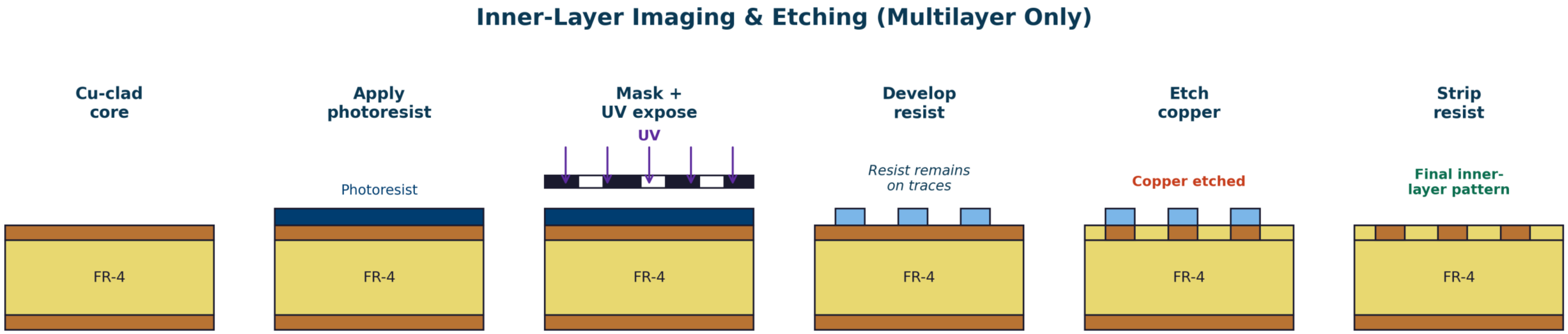


Why two steps?

(a) Electroless copper deposits ~1 µm seed everywhere — including non-conductive hole walls. (b) Electrolytic copper plating builds up to ~25 µm thickness using DC current — needs the seed to conduct.

Steps 7-8 — Outer-Layer Imaging + Etching

Same photolithography as inner layers — now on the outside copper

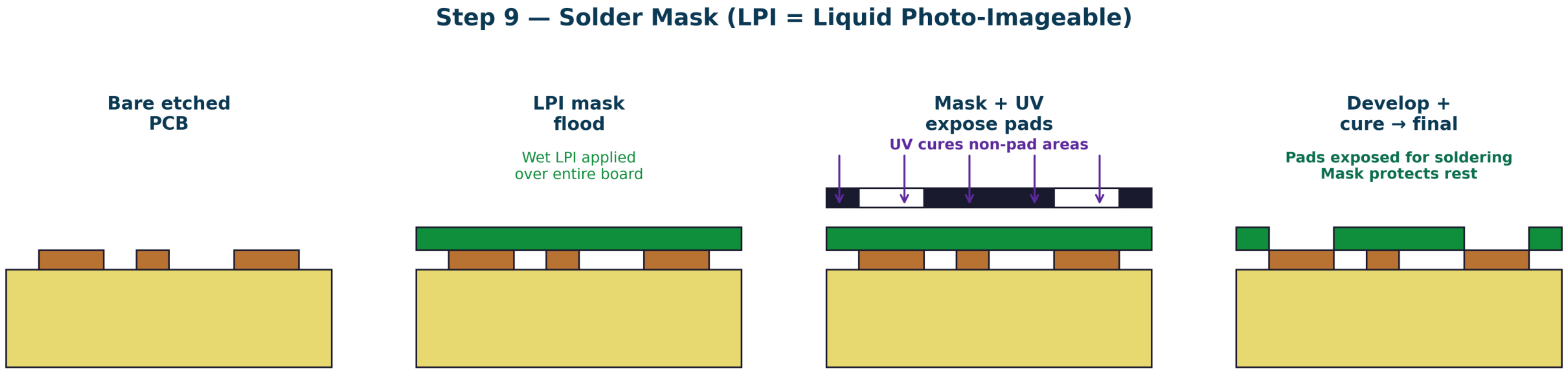


Difference from inner layers

Outer-layer etch must protect PTH holes — uses 'pattern plating' or 'tenting' to prevent the copper plated inside the holes from being etched away. Diagram pattern is the same as steps 2-3, just applied to outer Cu instead of inner.

Step 9 — Solder Mask

LPI (Liquid Photo-Imageable) green/black/red protective layer



Why solder mask matters

Protects copper from oxidation • Prevents solder bridges during assembly • Defines pad openings precisely • Improves visual readability of silkscreen above • Comes in green (cheapest), black (cosmetic), red, blue, yellow, white, purple

Step 10 — Silkscreen

Reference designators, logos, polarity marks — printed legend

- Epoxy-based ink applied through a screen or directly by inkjet
- Default colour: white on green board / black on white board
- Used for: ref-des (R1, C1, U1), polarity marks, logos, board version
- Minimum legible text: ~0.8 mm (32 mil) height
- Avoid silk over pads (gets covered by solder + looks unprofessional)
- DRC rule: 'silkscreen-on-pad' violation

Modern alternatives

Direct inkjet silkscreen — sharper, faster, more colors
Laser engraving — for high-end, no ink wear
QR codes printed in silkscreen — for serial-number tracking

Cost

Silkscreen is essentially free — included in standard PCB price.
Adding silkscreen on BOTH sides has no real cost impact.
Cosmetic / branding-grade silk is a slight upcharge.

Step 11 — Surface Finish

Protect exposed copper • Make pads solderable

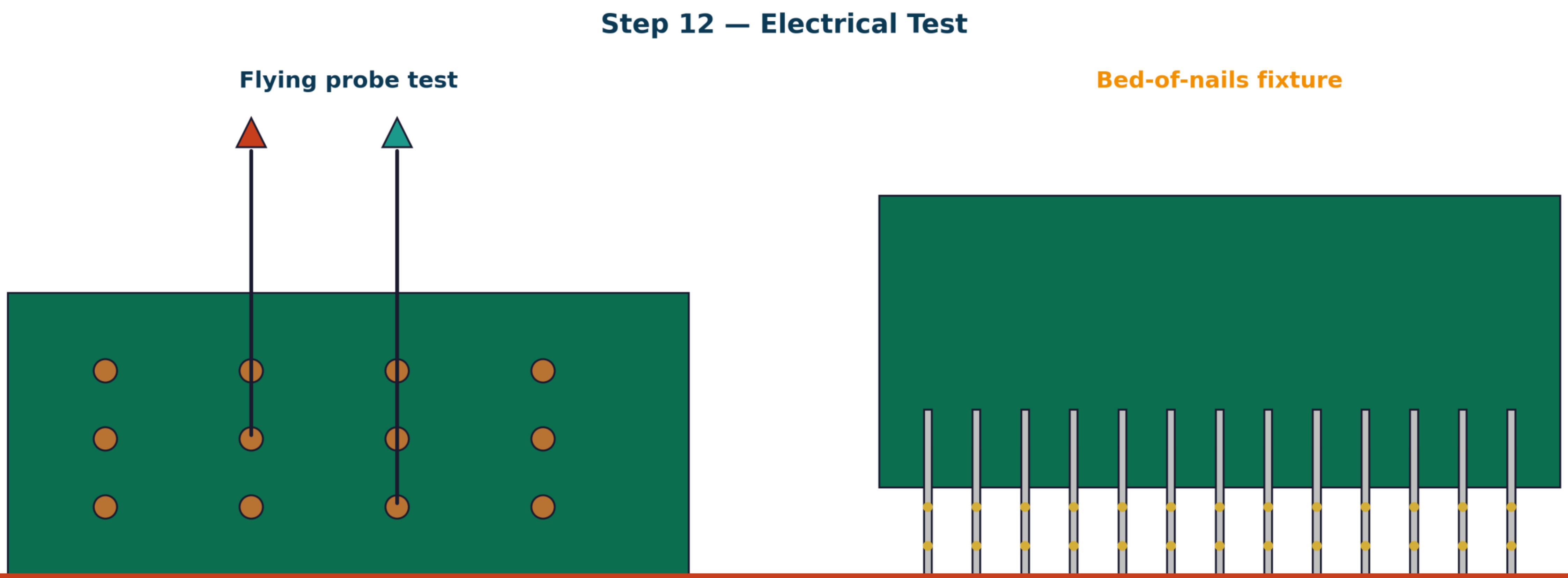
Step 11 — Surface Finishes

HASL	Lead-free HASL	ENIG
Hot Air Solder Level Sn-Pb dip + air-knife	SAC305 alloy RoHS-friendly	Electroless Ni + Immersion Au Gold finish
₹ *Cheapest, uneven*	₹₹ *Uneven, hot process*	₹₹₹ *Flat, best for BGA*

OSP	Immersion Ag	Hard Gold
Organic Solderability Preservative Clear film	Pure Ag on Cu Flat finish	Electrolytic Au Thick wear-resistant
₹ *Flat, short shelf life*	₹₹ *Flat, tarnishes*	₹₹₹₹ *For contact pads*

Step 12 — Electrical Test

Flying probe (low-vol) vs Bed-of-nails (high-vol)



What's being tested?

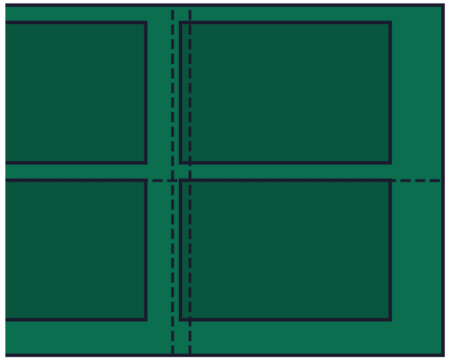
Continuity — every net has zero resistance from end to end • Isolation — adjacent nets have infinite resistance • Hi-pot — no shorts under 500-1500 V DC stress • Result: PASS or FAIL per board; fabs reject failed boards before shipping

Step 13 — Routing & Panel Separation

Cutting individual boards from the manufacturing panel

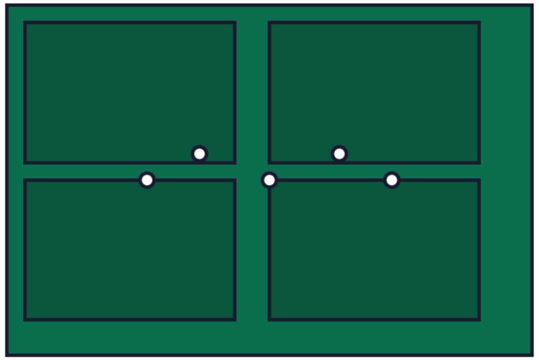
Step 13 — Routing & Panel Separation

V-groove



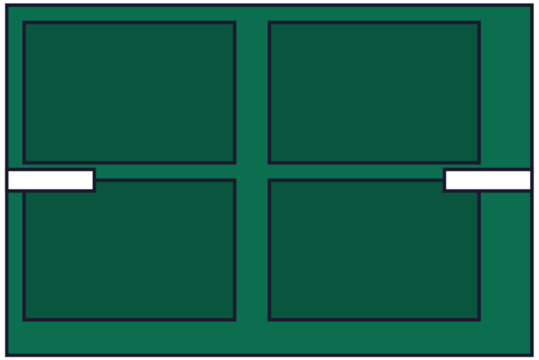
V-cut grooves on top + bottom
Snap apart after assembly

Mouse-bite tabs



Small drilled tabs hold boards together
Break off after assembly

Routed slots + tabs



CNC slot defines outline
Tabs hold panel together

Final step before shipping. Boards may also be packaged in panels for SMT line assembly first.

Process Comparison

What's different between single / double / multilayer

Process Comparison — Single vs Double vs Multilayer

Stage	Single-layer	Double-layer	Multilayer
1. Materials	Cu-clad FR-4	Cu-clad FR-4	Cores + prepreg + Cu foils
2. Inner-layer imaging	✗ Skipped	✗ Skipped	✓ Required
3. Inner-layer etch	✗ Skipped	✗ Skipped	✓ Required
4. Lamination	✗ Skipped	✗ Skipped	✓ Required
5. Drilling	✓ Mech.	✓ Mech.	✓ Mech. + laser (HDI)
6. Plating	✗ No PTH	✓ PTH required	✓ PTH required
7-8. Outer image+etch	✓ One side only	✓ Both sides	✓ Both outer sides
9. Solder mask	✓ One side	✓ Both sides	✓ Both sides
10. Silkscreen	✓	✓	✓
11. Surface finish	✓	✓	✓
12. Electrical test	Simple	Net-list	Net-list (extensive)
13. Cutting	✓	✓	✓
Lead time	1-3 days	3-5 days	5-15 days

Common Manufacturing Defects

What to look for at incoming inspection

Common PCB Manufacturing Defects

⚠ Open trace	⚠ Short circuit	⚠ Drill breakout	⚠ Annular ring
Cu missing → no connection	Adjacent traces bridged	Hole drilled off-pad	Insufficient pad around drill
⚠ Solder mask misalign	**⚠ Lifted pad**	**⚠ Cu void**	**⚠ Layer mis-register**
Mask offset → exposed copper	Pad detached from substrate	Plating incomplete in PTH	Inner layers offset from outer

How to Specify a PCB Order

Every line on the fab quote form, explained

- Quantity: 5 / 10 / 50 / 100+ — drives per-unit cost
- Board size: e.g., 50 × 30 mm — affects panel utilisation
- Layer count: 1 / 2 / 4 / 6 / 8 — biggest cost driver
- Material: FR-4 standard (Tg 130 / 170 / 180)
- Thickness: 0.8 / 1.0 / 1.6 / 2.0 mm
- Outer copper weight: 1 oz (default) / 2 oz (heavy current)
- Solder mask color: green (default) / black / red / blue / white
- Silkscreen color: white (default) / black
- Surface finish: HASL / lead-free HASL / ENIG / OSP
- Minimum trace width: 6 mil (default) / 4 mil (premium)
- Minimum drill: 0.3 mm (default) / 0.15 mm (premium / HDI)
- Impedance control: Y/N (additional cost & spec sheet)
- Edge plating / castellated holes: Y/N
- Panelization: per fab template OR custom panel file

Choosing the Right PCB Type

Decision matrix by criterion

Choosing the Right PCB Type

Criterion	Single	Double	4-layer	6-layer+
Component density	Low	Medium	High	Very high
Signal speeds	DC, kHz	< 50 MHz	< 500 MHz	GHz
Power planes	✗	Shared	Dedicated GND	Dedicated GND + PWR
Signal integrity	Poor	OK	Good	Excellent
EMC performance	Poor	Fair	Good	Excellent
Cost (10 pcs, 50×50 mm)	₹500	₹1500	₹5000	₹15k+
Lead time (proto)	1-3 d	3-5 d	5-7 d	10-15 d
Typical use	Toys, kits	General-purpose	MCU + RF	HDI / dense BGAs

Where PCBs Are Made

Global landscape • Indian fabs gaining ground

China

JLCPCB, PCBWay, AllPCB, Seeed
5-7 day lead, very cheap
80% of global PCB volume

Taiwan

Unimicron, Compeq, Zhen Ding
High-end (HDI, IC substrates)
Fastest tech advancement

Korea

Samsung Electro-Mechanics, LG
HDI, flex, advanced packaging

Japan

Ibiden, Nippon Mektron
Ultra-high-end (substrate-like)

USA / EU

Sanmina, Royal Circuit
Higher cost — for ITAR, medical, aero, low-volume

India

Centum, AT&S India, Genus, Epitome
Growing under PLI scheme
Good for proto / small-medium volume

Cost & Lead-Time Guide

What to expect from low-volume prototypes (5-50 pieces)

Spec	Single-layer	Double-layer	4-layer	6-layer
10 pcs, 50×50mm	₹500-1000	₹1000-2000	₹4000-8000	₹15000-25000
50 pcs, 50×50mm	₹1500-2500	₹2500-5000	₹10000-20000	₹30000-60000
Lead time (proto)	1-3 days	3-5 days	5-10 days	10-15 days
Min trace width	8 mil	6 mil	5 mil	4 mil
Min drill	0.4 mm	0.3 mm	0.25 mm	0.2 mm + microvia
Best for	Power, LED	Most products	SI / EMC focus	Dense BGA / RF

Volume break-points

Costs drop steeply between 50 → 500 pcs and again at 5000+ pcs. Real production runs (10k+) are ~10× cheaper per board than prototypes.

Conclusion

Key takeaways from PCB manufacturing

13 stages, predictable

Every PCB follows the same script.
Know the script, design for it.

Layer count drives cost

1L → 2L → 4L → 6L doubles cost each step.
Pick what you need, no more.

Multilayer adds 3 stages

Inner imaging + etch + lamination
are unique to multilayer.

Surface finish matters

HASL for cheap, ENIG for fine pitch, OSP for cost + flat.

Tooling is the cost driver

Per-board cost drops 10× from
prototype to mass production.

DFM saves rev cycles

Designing within fab capabilities = cheaper boards +
faster turnaround.

Questions & Discussion

Ask anything — process steps, vendor selection, defect identification, cost estimation.

References & Further Reading

- IPC-6012 — Qualification & Performance Specification for Rigid PCBs
- IPC-A-600 — Acceptability of Printed Boards (the bible of PCB inspection)
- IPC-2221 — Generic Standard on Printed Board Design
- JLCPCB / PCBWay knowledge bases — practical fab-side perspective
- Sierra Circuits — design + manufacturing how-to articles
- Phil's Lab YouTube — PCB factory tour + DFM walkthroughs
- MK Publications — Printed Circuits Handbook (Coombs)
- Indian PCB Association (IPCA) — local fab capabilities directory

Thank you • Visit a PCB fab if you can — invaluable experience